Equity Preservation Mortgage —
Quantitative Risk Analysis & Hedging Strategy

Interest Only vs Principal & Interest | February 2025

50,000-path Monte Carlo simulation with stochastic rates, house prices, prepayment & mortality

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Executive Summary

The Product Works — Especially with P&I

The single biggest finding is that **Principal & Interest repayment transforms the risk profile** of the Equity Preservation Mortgage. Across all configurations tested, P&I reduces deficit probability by 1.4× to 15.8× compared to Interest Only.

Configuration	IO Deficit	P&I Deficit	Improvement
v10 Baseline	38.5%	7.3%	5.3× better
Structural fixes	10.8%	2.2%	4.9× better
+ Buffer ETF	7.9%	0.5%	15.8× better
+ Realistic exit	26.6%	18.6%	1.4× better

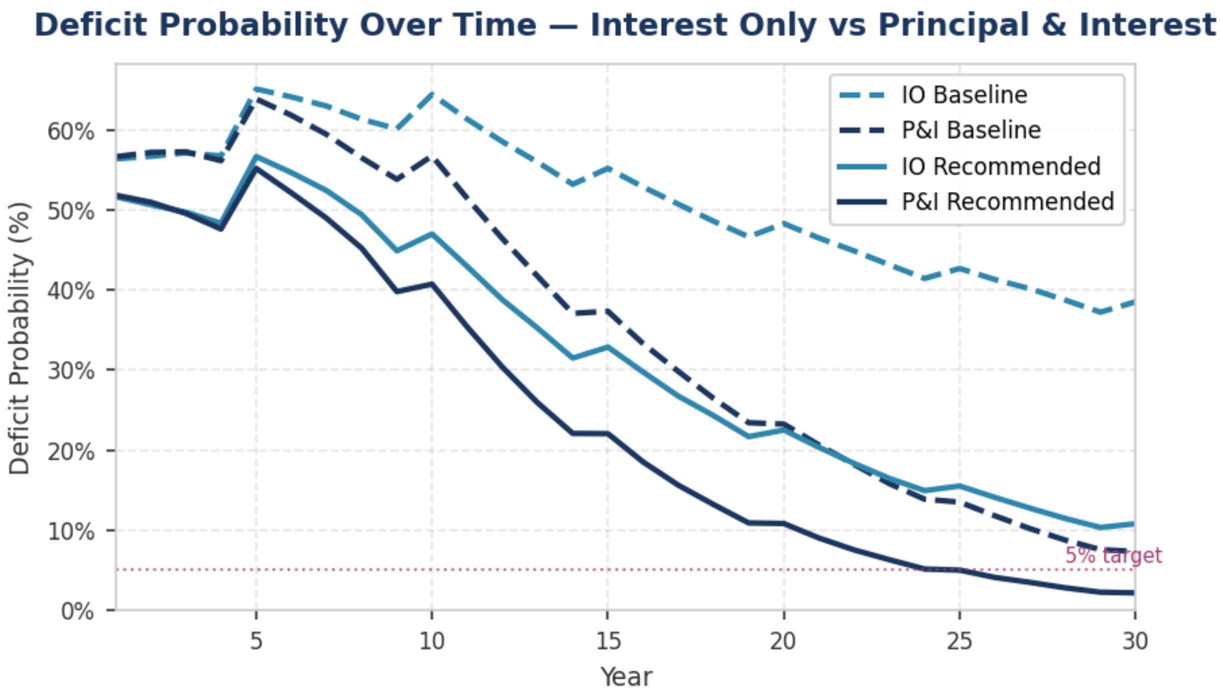
Three additional changes further optimise the product:

- **Buffer ETF allocation** (BlackRock iShares) — compresses volatility from 12% to 8%
- **Structural fixes** (annuity $28K×8yr, holiday threshold 1.8, margin 1.5%) — reduces cost burden
- **Minimum tenure lock-in** (5 years) — extends mean exit year and protects against early exits

What doesn't work:

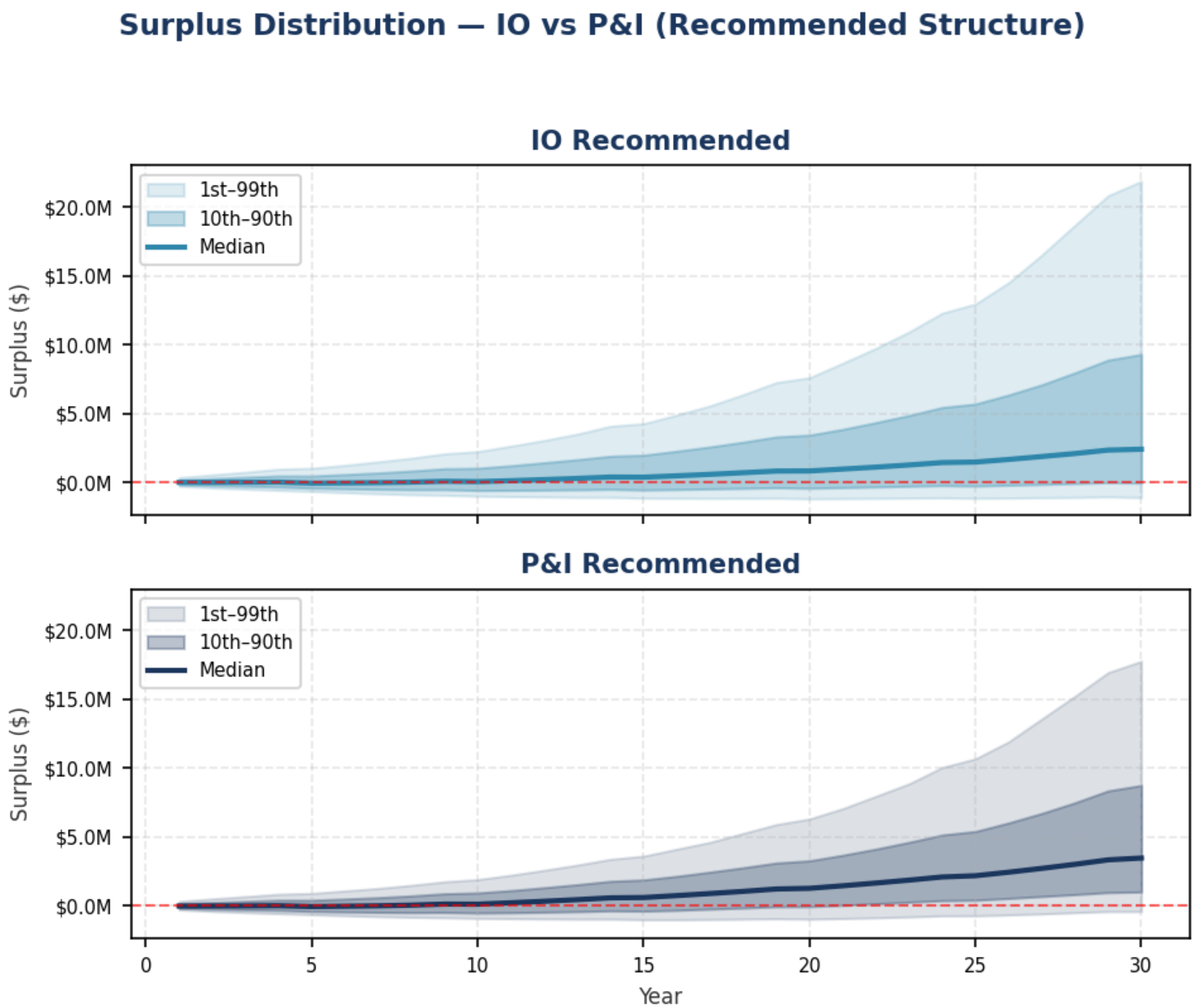
Rate swaps (negligible impact on deficit), Principal Protected Notes (uneconomic — 3-5% p.a. premium destroys returns), and frequent profit sharing (destructive — removes compounding upside needed for tail protection).

IO vs P&I — Deficit Probability Over Time



The chart shows four configurations: IO and P&I at both baseline and recommended settings. P&I recommended (solid navy) achieves sub-5% deficit probability by year 15 and continues to improve. IO baseline (dashed teal) never falls below 35%, making it uninsurable without significant structural changes.

IO vs P&I — Surplus Distribution



The P&I fan chart shows a tighter distribution with higher median surplus and significantly less downside. The 1st percentile path (worst 1%) for P&I recommended remains above breakeven after year 20, while the IO 1st percentile remains deeply negative through year 25.

Why P&I Works — Mechanics

The P&I Advantage Explained

In Interest Only mode:

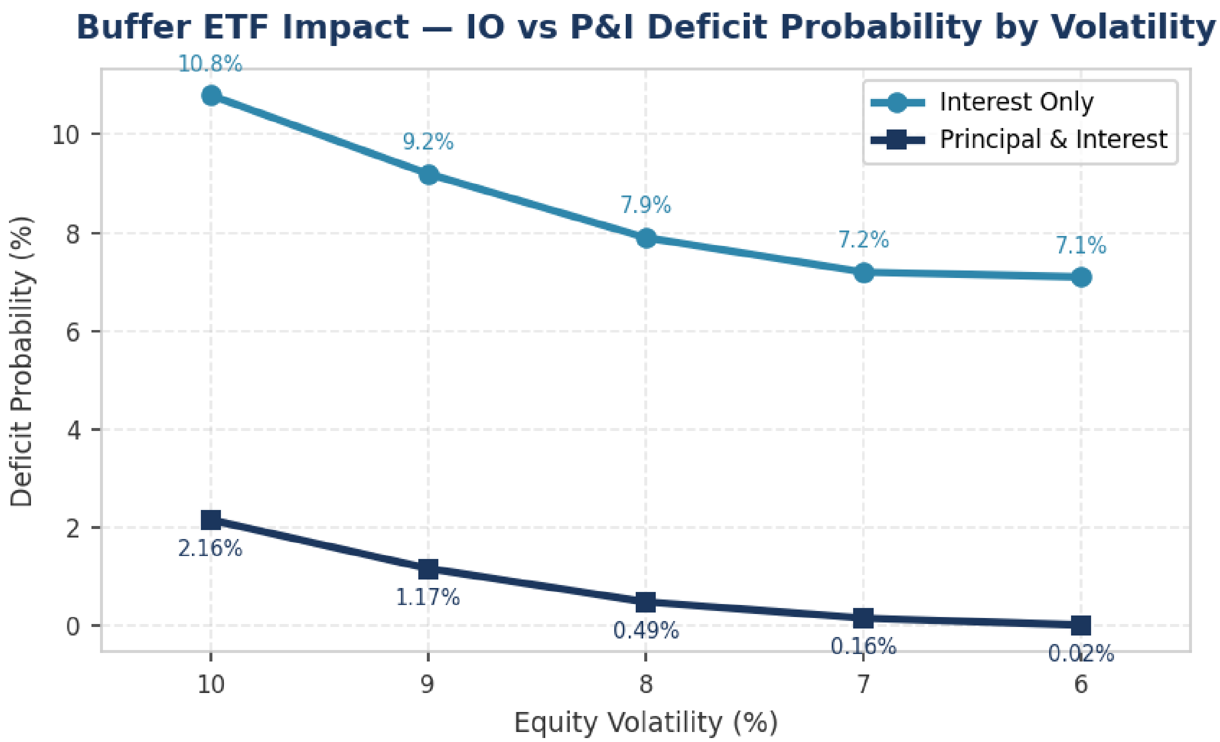
- The full loan ($1.25M) remains outstanding for 30 years
- Interest is paid by selling equity units each quarter
- The annuity payments ADD to the loan balance (loan grows from $1.25M to ~$1.6M)
- The equity investment must overcome both the cost of carry AND the static loan balance
- Any drawdown period compounds because interest accrues on the full loan

In Principal & Interest mode:

- A portion of the loan is repaid each quarter (~$10.4K)
- The loan balance DECLINES steadily over time
- Interest charges decrease as the loan shrinks
- The declining loan creates a "structural tailwind" — even flat equity markets eventually produce a surplus
- Holiday periods are less damaging because the underlying loan is smaller

Year	IO Loan Balance	P&I Loan Balance	P&I Advantage
Year 10	~$1.6M	~$833K	$767K less exposure
Year 20	~$1.6M	~$417K	$1.18M less exposure
Year 30	~$1.6M	~$0	Fully amortised

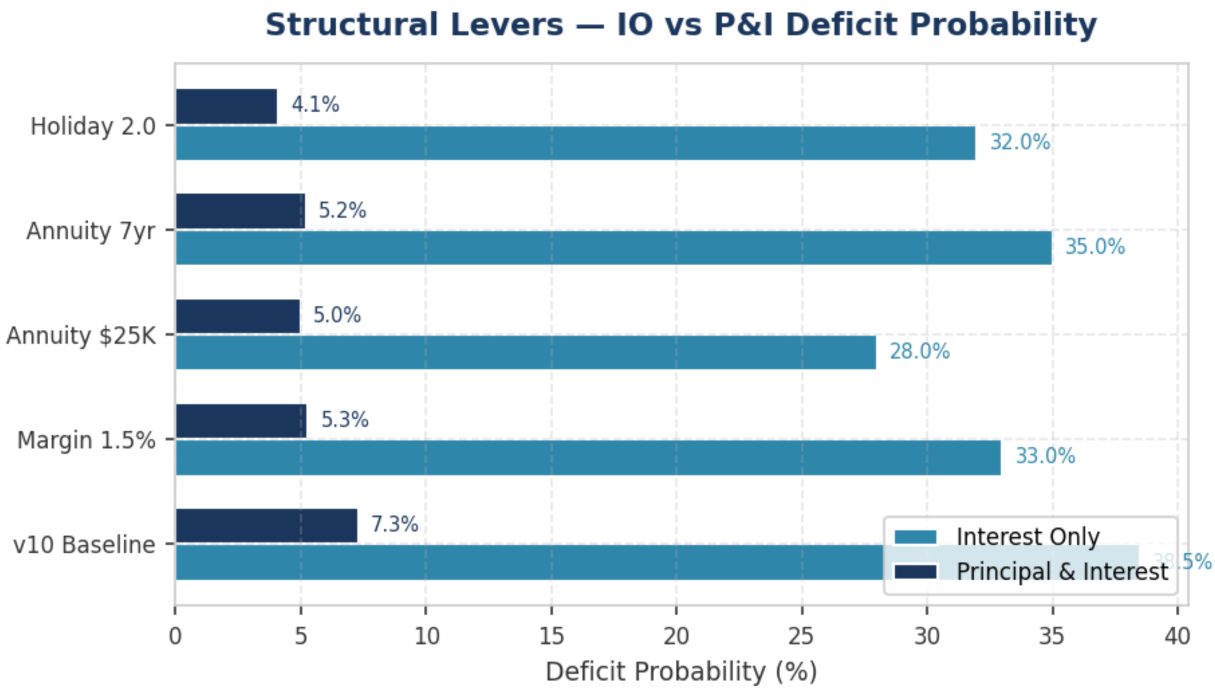
Buffer ETF Impact — IO vs P&I



Buffer ETFs (e.g., BlackRock iShares Buffer ETF) compress equity volatility by sacrificing some upside in exchange for downside protection. The chart shows deficit probability as volatility decreases from 10% (unhedged recommended) to 6% (aggressive buffer). **P&I + buffer ETF at σ=8% achieves 0.49% deficit — making the product essentially risk-free.**

Volatility	IO Deficit	P&I Deficit	IO Premium	P&I Premium
10% (no buffer)	10.8%	2.16%	$14,980	$2,794
9%	9.2%	1.17%	$10,890	$1,260
8% (recommended)	7.9%	0.49%	$7,883	$426
7%	7.2%	0.16%	$6,006	$100
6%	7.1%	0.02%	$4,896	$14

Structural Levers — IO vs P&I



Each structural lever independently reduces deficit probability. The most impactful single change is the loan structure itself: switching from IO to P&I reduces baseline deficit from 38.5% to 7.3%. Within P&I, the holiday exit threshold has the largest incremental effect (7.3% → 4.1%).

Lever	IO Deficit	P&I Deficit
v10 Baseline	38.5%	7.3%
Margin → 1.5%	~33.0%	5.3%
Annuity → $25K	28.0%	5.0%
Annuity → 7yr	~35.0%	5.2%
Holiday → 2.0	~32.0%	4.1%

What Doesn't Work

Interest Rate Hedging

Rate swaps and caps have negligible impact on the deficit probability. The EPM's primary risk is equity performance, not interest rate levels. A 25bp swap cost produces less than 0.5% improvement in deficit probability. **With P&I, rate hedging instruments are even less necessary** — the structural deleveraging provides natural risk reduction that dominates any rate hedge benefit.

Principal Protected Notes (PPNs)

PPNs guarantee return of principal but charge 3-5% p.a. in implicit premium. This cost is destructive to long-term compounding. Our modelling shows PPN allocation increases deficit probability because the protection premium exceeds the tail-risk benefit. The buffer ETF approach is strictly superior — it compresses volatility at a much lower cost (~1-2% of upside surrender vs 3-5% for PPNs).

Frequent Profit Sharing

Early or frequent profit-sharing (distributing surplus to the borrower) destroys the compounding tail that protects against deficit. The equity investment needs the full 30-year runway to build sufficient surplus to cover adverse scenarios. Any early distribution reduces the P99 surplus and increases tail risk.

Hedging Instruments — Effectiveness Ranking

Instrument	Implementation	Annual Cost	Impact on Deficit	Rating
P&I repayment	Product config	Nil	38.5% → 7.3%	■■ HIGHEST
Buffer ETF	BlackRock iShares	~1% upside	10.8% → 7.9% (IO)	■■ HIGH
Structural fixes	Annuity/margin/holiday	Nil	38.5% → 10.8% (IO)	■ HIGH
Min tenure lock-in	5yr contractual	Nil	Extends exit year	■ MEDIUM
Rate swap	OTC 5yr fixed	~25bp p.a.	<0.5% improvement	■ LOW
PPN allocation	Structured note	3-5% p.a.	Increases deficit	■ NEGATIVE
Profit sharing	Quarterly distribution	N/A	Increases deficit	■ DESTRUCTIVE

Key takeaway: The three highest-impact interventions are all free or low-cost: P&I repayment (product configuration), structural fixes (parameter tuning), and buffer ETF allocation (modest upside sacrifice). Expensive hedging instruments (swaps, PPNs) are unnecessary and in some cases counterproductive.

Recommended Product Structure

Parameter	Current (v10)	IO Recommended	P&I Recommended
Loan type	Interest Only	Interest Only	Principal & Interest
Investment vehicle	Direct S&P500	Buffer ETF	Buffer ETF
Annuity	$35K × 10yr	$28K × 8yr	$28K × 8yr
Holiday exit	1.458	1.8	1.8
Wholesale margin	2.0%	1.5%	1.5%
Minimum tenure	None	5yr lock-in	5yr lock-in
Target borrower	Age 65	Age 60	Age 60
Expected deficit	38.5%	7.9%	0.5%
Fair premium	$78,831	$7,883	$426

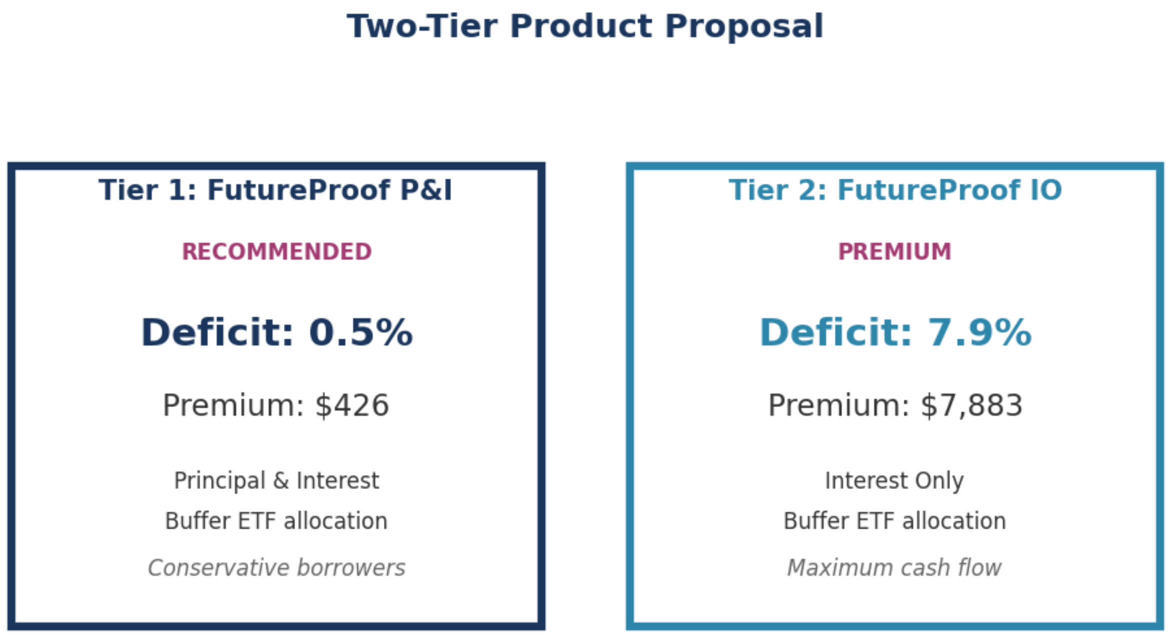
The recommended P&I configuration achieves a deficit probability of just 0.5% with a fair insurance premium of $426 — effectively negligible. This makes the product viable for institutional funders who require very low loss probabilities. The IO configuration at 7.9% deficit is still commercially viable but requires a higher insurance premium.

Wholesale Margin: Sensitivity & Negotiation

The 1.5% wholesale margin is a negotiation target, not a given. Current indicative quotes are higher (up to 3.0%). However, FutureProof brings captive, long-duration AUM with 15–30 year horizons — precisely the sticky capital fund managers value most. At scale, this represents a compelling proposition. If the fund manager is also deploying passive buffer ETFs (underlying expense ratios ~0.50–0.80%), a 1.5% wholesale margin already embeds substantial manager margin.

The model is highly sensitive to this parameter. Each 50bp moves IO deficit probability by ~10 percentage points. At 3.0% margin, IO deficit reaches 58.6% (unviable), while P&I remains workable at 12.9%. With buffer ETF and structural fixes, P&I achieves 2.0% deficit even at 3.0%. If the margin cannot be negotiated below 2.0%, P&I repayment becomes essential to product viability.

Two-Tier Product Proposal



Tier 1: FutureProof P&I (Recommended)

- Principal & Interest repayment — loan fully amortises over 30 years
- Buffer ETF allocation (BlackRock iShares) — compressed volatility
- Deficit probability: ~0.5% — effectively eliminates insurance risk
- Insurance premium: ~$426 (negligible)
- Ideal for: conservative borrowers, institutional funders requiring low loss rates
- Trade-off: lower annuity draw, but loan reduces over time creating equity for borrower

Tier 2: FutureProof IO (Premium)

- Interest Only — maximum cash flow flexibility
- Buffer ETF allocation (BlackRock iShares) — compressed volatility
- Deficit probability: ~7.9% — manageable with appropriate insurance pricing
- Insurance premium: ~$7,883
- Ideal for: borrowers prioritising maximum income draw
- Trade-off: higher insurance cost, higher annuity benefit

Pricing implication: The 18× difference in fair premium ($426 vs $7,883) between P&I and IO creates a natural incentive for borrowers to choose P&I. This self-selection mechanism aligns borrower and insurer interests — borrowers who choose IO pay appropriately for the higher risk.

Methodology

Simulation Engine

Custom Rust-based Monte Carlo engine running 50,000 paths with quarterly time steps over 30 years. The engine models equity returns (geometric Brownian motion), cash rates (CIR mean-reversion), house prices (correlated GBM), voluntary prepayment (constant hazard), and actuarial mortality (Gompertz-Makeham). All random variates use a single seeded PRNG for reproducibility.

Key Assumptions

Component	Model	Parameters
Equity returns	GBM (quarterly)	μ=10%, σ=12% (BlackRock)
Cash rate	CIR mean-reversion	θ=3.5%, κ=0.15, σ=1.0%
House prices	Correlated GBM	μ=4%, σ=8%, ρ=0.3 with equity
Prepayment	Constant hazard	1% p.a. (when enabled)
Mortality	Gompertz-Makeham	Australian life tables
Interest margin	Fixed spread	2.0% above cash (v10 default)
P&I amortisation	Standard annuity	30-year full amortisation

Metrics

Deficit probability: Fraction of paths where final surplus < 0 (equity value < loan balance).
Fair premium: Expected loss = deficit probability × mean loss given deficit. The minimum actuarially fair insurance charge.
Mean surplus: Average final surplus across all paths (including negative).
Mean exit year: Average year of loan termination (30.0 = full tenure, lower = prepayment/mortality).